

Q12

Permutation in string

$S_1 = "ab"$

$S_2 = "eidbaooo"$

By sliding window approach:-

$S_2 = "eidbaooo"$

1st window $\neq S_1$
remove add

$S_2 = "eidbaooo"$

2nd window

$\neq S_1$

remove add

$S_2 = "eidbaooo"$

3rd window $\neq S_1$

remove

add

$S_2 = "eidbaooo"$

4th window $= S_1$

return True

Code 1:-

//

```
class soln {
    public static void main (String args []) {
        int n = s1.length();
        int m = s2.length();

        if (m < n) {
            return false;
        }

        int [] s1count = new int [26];
        int [] windowcount = new int [26];

        for (int i = 0; i < n; i++) { // freq. of s1.
            s1count [s1.charAt(i) - 'a']++;
            windowCount [s2.charAt(i) - 'a']++;
        }

        if (matches (s1count, windowCount)) {
            return true; // First window check.
        }

        for (int i = n; i < m; i++) { // Sliding window.
            windowCount [s2.charAt(i) - 'a']++;
            windowCount [s2.charAt(i-n) - 'a']--;

            if (matches (s1count, windowCount)) {
                return true;
            }
        }

        return false;
    }
}
```

↓
compare.

add new character
remove old character


```
public boolean matches (int[] a, int[] b) {  
    for (int i = 0; i < 26; i++) {  
        if (a[i] != b[i]) {  
            return false;  
        }  
    }  
    return true;  
}
```

* Time Complexity :- $O(n)$